

SERUM INSTITUTE OF INDIA: COVID-19 VACCINE PRICING¹

Malay Krishna and Sunny Arora wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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I am happy to announce that the Indian vaccine industry will dedicate half of its COVID-19 vaccine production capacity – of billions of doses – to low- and middle-income countries, served by GAVI,² at an affordable and accessible price for all. Because your survival should not be determined by where you live in the world.

– Adar Poonawala, chief executive officer of the Serum Institute of India, June 2020³

It was 80 degrees Fahrenheit in the shade – typical for early August in the city of Pune, India – and a fine bead of sweat trickled down the brow of Adar Poonawala, chief executive officer (CEO) of the Serum Institute of India (SII). However, Poonawala was in one of the SII's temperature-controlled production facilities; the heat he felt was the result of concern about the pricing decision he needed to make. It was August 6, 2020, and the pricing strategy for the SII's pipeline COVID-19 vaccine had to be revisited. The SII was the world's largest manufacturer of vaccines. Indeed, it had produced over 1.5 billion doses in 2019.⁴ Following the characterization of the COVID-19 outbreak as a deadly global pandemic on March 11, 2020, the SII had entered the race to produce a vaccine. Just two weeks earlier, Poonawala had made a public announcement that thanks to a licensing agreement with AstraZeneca, the British-Swiss pharmaceutical giant, the SII would begin the manufacturing of the Oxford vaccine (one of the world's most promising vaccine candidates) in August 2020.⁵ If approved, the vaccine would be released for distribution in India as early as December 2020. It would be available at a price of US\$13⁶ per dose under the brand name "Covishield."⁷ Two doses of the vaccine would be required, resulting in a total cost of \$26 per person.

The price of \$13 per dose was much higher than the price of the SII's typical vaccine. Poonawala's father, Cyrus, the founder president of the SII, had boasted that for a dose of the measles⁸ vaccine, the SII price was just 50 cents, while Western big-pharma charged \$15.⁹ In the announcement about the agreement with AstraZeneca, Poonawala had explained the rationale behind the \$13 price: the SII had invested \$100 million in a new facility capable of producing one billion doses of COVID-19 vaccines annually.¹⁰ Additionally, the SII had committed to producing 200 million doses of Covishield ahead of its regulatory approval, at the company's own risk, to ensure the timely delivery of the vaccine.¹¹ The operating expenses associated with the production of this early batch of vaccines amounted to more than \$100 million. Moreover, the prices charged by United States (US) vaccine makers were higher; for example, Pfizer had recently announced a \$19.50 per dose supply deal with the US government.¹² The cost of treatment with the antiviral drug Remdesvir was estimated to be more than \$3,000 per patient.¹³ Even the cost of a COVID-19 test was more

than \$20.¹⁴ Yet, less than 24 hours earlier, Johnson & Johnson (J&J), a US-based pharmaceutical and consumer healthcare giant, had announced an agreement to supply a COVID-19 vaccine to the US government for a “not-for-profit” price of \$10 per dose.¹⁵ In another surprising development, five days previously, Russian authorities had declared that they would be ready to mass-distribute a vaccine by early October 2020.¹⁶ Poonawala needed to rethink the SII’s pricing strategy for Covishield, and quickly.

COVID-19

COVID-19 was caused by a virus named SARS-CoV-2, which was highly infectious, had no known cure, and had increasingly serious effects with age.¹⁷ The World Health Organization (WHO) first became aware of the new disease through Chinese media reports in December 2019.¹⁸ By March of 2020, it was clear that the new virus was both deadly and widespread – leading to it being declared a global pandemic.¹⁹ In the absence of a cure or vaccine, the only protections available for public health authorities to recommend were the tried-and-tested guidelines of social distancing, mask wearing (as the virus was transmitted through airborne droplets), and practicing good hand hygiene.²⁰ The political and public responses to these guidelines varied by country, state, and even city.²¹ Several large countries, including Brazil, India, and the US, continued to struggle with steady increases in their COVID-19 infection rates.²² While some countries were reported to have successfully contained the spread of the virus by June 2020, later surges led to restrictions on public movement even in countries like Australia, Switzerland, Germany, and South Korea.²³ A surge in cases could quickly overwhelm a country’s health infrastructure. It was estimated that 1 per cent of infected people in their twenties, 8 per cent of people in their fifties, and 19 per cent of infected people over 80 years of age needed to be hospitalized.²⁴ These problems were exacerbated by the insufficient capacity of testing centres.²⁵

The public health crisis also had economic consequences worldwide. In fact, the International Monetary Fund (IMF) predicted a global economic contraction of 4.9 per cent in 2020, with an uncertain recovery in 2021.²⁶ The economic contraction fostered pressure to reopen commercial and public spaces, although each relaxation of restrictions resulted in surging infections, hospitalizations, and deaths. While new therapies reduced death rates, a cure for the disease remained elusive.²⁷ Thus, hopes of returning to normalcy depended on the access to safe and effective vaccines that could protect against the severe effects of the disease. Vaccine development accelerated to an unprecedented extent, driven by innovations in research and development and cooperation across the public-private ecosystem of vaccine development.²⁸ Some experts expressed concern that a vaccine development process that usually took several years had been shortened to a matter of months.²⁹ Experts also warned that the first vaccines might not confer long-term protection for the population.³⁰ However, the increased speed of vaccine development was accepted as necessary, driven by advances in technology and fast-tracking by public health officials.³¹

VACCINE BUYERS

Countries worldwide wanted to acquire a sufficient stock of vaccines for widespread distribution at no cost or little cost to consumers. However, the ability to purchase vaccines depended on their price as well as on the countries’ purchasing power. A commonly used proxy for purchasing power was a country’s per capita gross national income (GNI), which formed the basis of the World Bank’s classification of countries into four categories—namely high income, upper-middle income, lower-middle income, and low income. The high-income countries had an average per capita GNI of \$46,000, and they were home to 1.2 billion of the world’s population in 2019 (see Exhibit 1). The US, the most populous high-income country, did not regulate vaccine pricing, and early reports from the US speculated that a COVID-19 vaccine could cost hundreds of dollars or more.³² By the end of June 2020, signals from vaccine makers indicated a much

lower price. For instance, Moderna reassured the public that the company's prices would be in line with existing vaccine prices in the US.³³ On July 22, Pfizer announced a deal to supply the US government with its COVID-19 vaccine (if approved) at a price of \$19.50 per dose.³⁴ This was at the lower end of vaccine prices in the US (see Exhibit 2A).

Further, J&J and the UK-based AstraZeneca had promised the US a not-for-profit supply of vaccines during the pandemic.³⁵ However, none of these new vaccines had yet passed the clinical trial stage. Research indicated that the probability of success for vaccines at each step of a clinical trial was 76 per cent (for phase one), 57 per cent (for phase two), and 85 per cent (for phase three).³⁶ To improve its chances of acquiring an effective and approved vaccine, the US had signed contracts with several companies that were leading the race to produce a vaccine.³⁷ These contracts were contingent on the vaccine clearing phase-three clinical trials. If multiple vaccines achieved regulatory approval, the final bill for taxpayers could exceed several hundred dollars per person. Similar arrangements were in place in other high-income countries. For example, the UK, which had an estimated population of 66 million, had contracted with AstraZeneca for 100 million doses, with the French company Valneva for 60 million doses, and with Pfizer for 30 million doses.³⁸

The upper-middle-income countries, which were home to 2.9 billion people, had an average per capita GNI of \$9,000 in 2019. In China, the most populous country in that category, the discretionary³⁹ part of the national budget was estimated to be \$2,200 per capita (see Exhibit 1). As China's population exceeded 1.4 billion, the issue of mass supply was critical. The production capacity of China's vaccine industry was estimated to be one billion doses per year, and it had achieved a production volume of 0.7 billion doses in 2019.⁴⁰ Three Chinese candidate vaccines had entered or were soon to enter phase-three clinical trials as of early August.⁴¹ Most of China's routine vaccines were produced by Chinese companies, distributed through public vaccination programs, and priced between \$0.10 and \$6, which was similar to UNICEF's⁴² pricing (see Exhibit 2B).⁴³ However, the prices for private-market vaccines in China could be as high as \$100, which was a higher amount than prices for comparable US, European, and UNICEF vaccines.⁴⁴

In the lower-middle- and low-income countries, vaccine prices were typically 10 per cent of the price in the US and Europe.⁴⁵ The lower-middle-income nations, where 2.7 billion people resided, had an average per capita GNI of \$2,200 in 2019. India, which had a population of 1.3 billion, was the most populous of those countries, and its discretionary budget outlay was \$200 per capita. Thus, a two-shot course of the SII's Covishield vaccine for India's population could represent a significant budgetary allocation. In 2019, India's per capita budget for healthcare was less than \$10.⁴⁶ However, India was also the world's fifth largest economy, with an investment-grade national debt rating, which meant it could likely finance a mass vaccination program.⁴⁷ India's vaccine production capacity was the largest in the world, standing at three billion doses in 2019, of which an estimated two billion doses were exported.⁴⁸ Aside from the SII, two other Indian vaccine makers – Bharat Biotech and Zydus Cadila – were developing COVID-19 vaccines, which entered early-stage human trials in August 2020.⁴⁹ If approved, they would each produce approximately 100 million doses per year, beginning in the first quarter of 2021.⁵⁰ The prices were expected to be low. The low-income countries were home to 0.6 billion people in 2019, and the discretionary budget per person for a representative country, such as Ethiopia, was \$70 (see Exhibit 1). In June 2020, a deal to supply several hundred million doses of affordable vaccines to low- and lower-middle-income countries was announced.⁵¹ The deal included multiple partners, such as the Gates Foundation; the Coalition for Epidemic Preparedness Innovations (CEPI); GAVI, the Vaccine Alliance; AstraZeneca; and the SII.⁵²

THE COMPETITIVE DYNAMIC

By early August 2020, six vaccines had entered final-phase clinical trials, with candidates having been developed by Pfizer-BioNTech, AstraZeneca-Oxford University, Moderna, Sinopharm, and Sinovac.⁵³ A few other vaccines, such as those developed by J&J and Russia's Gamaleya Institute, were expected to enter phase-three trials by the end of September 2020.⁵⁴

US-based Pfizer, one of the world's largest pharmaceutical companies, had announced a deal to supply the US government with 100 million doses of the vaccine developed by its German partner, BioNTech, for \$1.95 billion.⁵⁵ According to the arrangement, the US government would receive the first batch of doses by the end of 2020, and it would have the right to acquire up to 1.3 billion more doses by the end of 2021. Pfizer had also announced similar deals with the UK (30 million doses) and Japan (120 million doses).⁵⁶ During an earnings-related conference call, the senior management of Pfizer had explained the logic behind the vaccine's pricing:⁵⁷ during the pandemic, the vaccine would be priced to allow for "broad access" and supplied to multiple governments around the world. Pfizer believed that the true value of the vaccine was far greater than the initial price of \$19.50. After the pandemic, Pfizer expected the demand for vaccines to "become more standard or seasonal," leading to normal channels of supply and a more "value-based" pricing approach, depending on both the competition and the demand at the time. Pfizer also expected that due to the huge demand – given that the world's population stood at 7.5 billion – competition would not erode price levels in the near future. Pfizer expected its vaccine to be one of the earliest to reach the market in late 2020, partly because it had declined US government funding, thereby avoiding lengthy negotiations regarding pricing.⁵⁸ Moreover, Pfizer had received approval from the US regulator (the Food and Drug Administration) to fast track its clinical trials by combining phases two and three.⁵⁹

In contrast with the other key vaccine manufacturers, Moderna was a new biotechnology venture. It was established in the US in 2010, and it was only listed as a publicly traded company in 2018.⁶⁰ While Moderna had developed a range of medicines and vaccines, all of its products were based on mRNA⁶¹ technology.⁶² The principal advantage of this technology was that it allowed products to have a much smaller footprint than conventional pharmaceutical and vaccine products.⁶³ The risk was that no mRNA vaccine had previously been approved anywhere in the world.⁶⁴ The Pfizer-BioNTech vaccine was also based on mRNA technology. Unlike Pfizer, however, Moderna had received \$955 million in US government funding to develop its vaccine.⁶⁵ Yet, despite accepting the funding, Moderna had insisted that it could not price the vaccine at cost, as it needed the vaccine sales to return profits for its investors.⁶⁶ During an earnings-related conference call on August 5, 2020, Moderna's CEO had also announced that the company had forged deals to supply vaccines to a number of European Union (EU) countries at a cost of \$32–\$37 per dose.⁶⁷ Attempting to strike a more reassuring note, he suggested that for deals involving a higher volume, the price would be lower. Moderna had entered into three agreements concerning the production, packaging, and storage of its vaccines. First, it had contracted with Lonza, a Switzerland-based multinational biotechnology corporation, to manufacture vaccine material equivalent to one billion doses per year.⁶⁸ Second, it had contracted with US-based Catalent to package and store 100 million doses of its vaccine in the US.⁶⁹ Third, it had contracted with Spain's ROVI to package and store hundreds of millions of vaccine doses in pan-European locations.⁷⁰

AstraZeneca was a vaccine frontrunner from Europe. In March 2020, AstraZeneca had signed a licensing agreement with Oxford University's Jenner Institute for the experimental COVID-19 vaccine developed at the institute.⁷¹ According to the agreement, both partners agreed to operate on a not-for-profit basis for the duration of the coronavirus pandemic; however, after the pandemic, both would receive royalties based on commercial pricing.⁷² The Oxford vaccine, as it was commonly known, stimulated the human body's production of antibodies and high levels of T-cells⁷³ specific to the COVID-19 virus. The head of the Jenner Institute had characterized this immune response as "clearly better" than Moderna's vaccine.⁷⁴ In May, the company announced that a US government agency had pledged funding support of up to \$1.2 billion for

the development, production, and delivery of the vaccine toward the end of 2020.⁷⁵ The development program included phase-three clinical trials involving 30,000 participants as well as a pediatric trial.⁷⁶ If the vaccine received approval, AstraZeneca would supply 300 million doses to the US and 100 million doses to the UK on a not-for-profit basis.⁷⁷ AstraZeneca had also announced an expanded deal with the EU to supply member states with 400 million doses.⁷⁸ After committing to expanding its global production to over one billion doses by the end of 2021, AstraZeneca signed licensing agreements across the world, including with the SII in India, Shenzhen Kangtai in China, and others in South Korea and Brazil.⁷⁹

On August 5, 2020, J&J announced a deal to supply the US government with 100 million doses of its vaccine in exchange for \$1 billion.⁸⁰ The arrangement also allowed the US government to order an additional 200 million doses from J&J.⁸¹ The agreed \$10 per dose price was lower than the price of most existing vaccines in the US,⁸² which aligned with J&J's pledge to provide a supply of its vaccine on a not-for-profit basis during the pandemic.⁸³ Unlike Pfizer, J&J had accepted \$450 million in US government aid to help accelerate the development of its COVID-19 vaccine.⁸⁴ Once the pandemic was over, J&J would be expected to shift to for-profit pricing.⁸⁵ J&J's vaccine was expected to enter phase-three clinical trials in late September 2020.⁸⁶

The two Chinese vaccine manufacturers that were in final-phase trials on August 5, 2020, namely Sinopharm and Sinovac, had based their vaccines on the older, tried-and-tested approach of inactivating the disease-causing virus.⁸⁷ Another Chinese vaccine manufacturer, CanSino, was expected to enter phase-three trials soon, and its candidate vaccine was based on more advanced genetic technology, similar to the Oxford and J&J vaccines.⁸⁸ All of the Chinese vaccine manufacturers faced one key issue: the difficulty of recruiting suitable human subjects for phase-three trials due to the absence of a COVID-19 epidemic⁸⁹ in China in August 2020. Thus, China had entered into agreements with countries that were facing epidemic conditions, such as Russia, Brazil, and Pakistan, to test its vaccines there.⁹⁰ Another issue concerned perceptions, as top US health officials doubted the quality of Chinese vaccines due to prior mishaps involving defective vaccines.⁹¹

The Russian vaccine was developed by the state-backed Gamaleya Institute. The Russian health minister had declared that mass vaccinations would begin, starting with medical workers and teachers, in early October 2020, which would make it the first mass vaccination for COVID-19.⁹² By August 5, it had become clear that the early phase human trial data were promising, and the Russian vaccine entered phase-three clinical trials.⁹³ The Russian health minister had also announced that three other vaccines were then under development, with one of the candidates having entered early-stage human trials.⁹⁴ A leading Russian virologist involved in the project claimed that the motive was neither profit nor politics, but instead purely humanistic. He also hinted that after Russian citizens had been vaccinated, other countries would be considered.⁹⁵ Early vaccine production in Russia was expected to reach 1.5 million doses per month, progressing to six million per month by the end of 2021.⁹⁶

Global vaccine production capacity was an overarching concern. The WHO had estimated the world's production capacity for all vaccines to be about 3.5 billion doses per year.⁹⁷ Reports from across the globe had indicated plans to more than double the vaccine production capacity so that more than four billion doses of COVID-19 vaccines could be produced by 2021.⁹⁸ However, even doubling the capacity would imply a global shortfall in supply in 2021, given that most vaccines required two doses.

SERUM INSTITUTE OF INDIA

The SII was founded by Cyrus Poonawala in 1966, a time when his home country of India had to import most of its vaccines at high prices. Based on domestic market requirements, the SII developed the know-how and technology necessary to manufacture several vaccines and related immunological products, such

as tetanus anti-toxins, at an affordable price.⁹⁹ Over time, the SII began to export its vaccines to many low- and middle-income countries, and eventually, to high-income countries. In 2019, the SII produced and sold more than 1.5 billion doses of vaccines and related biopharmaceutical products used in over 170 countries.¹⁰⁰ It was estimated that about two-thirds of the world's children had received at least one vaccine manufactured by the SII.¹⁰¹ These vaccines were accredited by the WHO, and they were favoured due to their quality and affordability.¹⁰²

Despite its policy of keeping vaccine prices low, the SII enjoyed profit margins of over 40 per cent, which were higher than the profit margins of its larger and more diversified, global competitors (see Exhibit 3). The top management of the SII attributed the high margins to the company's ability to keep overhead costs low, a motivated workforce, as well as advanced automation and production equipment.¹⁰³ Cyrus also believed that as the SII was privately held, it could take a long-term view regarding investments, which was harder to do in publicly traded firms. An example of this thinking could be seen in the SII's large, temperature-controlled warehouse, which maintained a considerable stock of a variety of vaccines. While the warehouse increased operating expenses, the ready inventory enabled the SII to respond rapidly to an outbreak.¹⁰⁴ A source of reduced expenses was the lack of investment in research concerning new vaccines. Instead, the SII focused its development efforts on improving production yields and increasing the stability of its vaccines across a range of temperatures.¹⁰⁵ As the SII did not develop its own vaccines, it worked with off-patent products, while for newer products, it entered into licensing agreements.¹⁰⁶ Thus, the SII had entered into agreements regarding COVID-19 vaccines with AstraZeneca and Novovax, another leading vaccine developer.¹⁰⁷ However, in a departure from this strategy, the SII had also invested in developing its own COVID-19 vaccine through modifying one of its existing, off-patent vaccines.¹⁰⁸

THE PRICING DECISION

Poonawala had announced that the SII would introduce the Covishield vaccine at an initial price of \$13 per dose, with a price reduction occurring over time as the SII realized economies of scale.¹⁰⁹ The initial \$13 price had seemed fair, given the higher prices of the Pfizer and Moderna vaccines, the Remdesvir drug, and even COVID-19 testing kits. Yet, the price point was under attack, given the information released by J&J and the Russians. Still, the investment in the new plant and the risk and operating expenses incurred through pre-approval production—amounting to more than \$200 million—had to be recovered.¹¹⁰ Should the SII continue to follow Pfizer's two-part pricing model of “broad access” during the pandemic followed by “value-based” access during the endemic phase (after the pandemic)? But what if the next generation of vaccines entered the market during the pandemic? How should the launch price take such uncertainty into account? Should the SII continue with the \$13 price after all?

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EXHIBIT 1: KEY CHARACTERISTICS OF COUNTRY SEGMENTS BY INCOME

Segment ^a →	High-income economies	Upper-middle-income economies	Lower-middle-income economies	Low-income economies
Characteristics ↓				
Gross national income (GNI) per capita criterion, ^a for inclusion in segment	> \$12,356	\$4,046 to \$12,535	\$1,036 to \$4,045	< \$1,035
Population in segment (billions)	1.2	2.9	2.7	0.6
Weighted average ^b GNI per capita	\$45,671	\$9,014	\$2,190	\$703
Representative (most populous) country	United States	China	India	Ethiopia
Per capita GNI of representative country, 2019	\$65,760	\$10,410	\$2,130	\$850
Per capita national budget outlay of representative country, 2019	\$13,468	\$3,832	\$306	\$133
Proportion of 2019 budget outlay considered discretionary ^c for representative country	30%	57%	64%	52%
Per capita discretionary national budget outlay of representative country, 2019	\$4,040	\$2,198	\$197	\$69
Typical ^d discretionary budget spend category (% of 2019 discretionary budget)	Veterans' benefits and services: 6.5%	Public security: 6.6%	Social welfare: 2.8%	Justice and security: 4.0%
Per capita discretionary healthcare budget of representative country, 2019	\$3,412	\$165	\$7	\$14

Notes: ^a According to the World Bank classification. "World Bank Country and Lending Groups," World Bank Data Help Desk, July 1, 2020, accessed September 24, 2020, <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

^b Weighted by population of countries belonging to the segment.

^c Excludes mandatory budget items such as national debt obligations, allocation for public pensions, government salaries, and other legally binding outlays.

^d Measure of the central tendency, median of all the discretionary spend categories for each representative country.

Source: Created by the case authors based on information from "GNI (Current US\$)," The World Bank | Data, accessed August 25, 2020, <https://data.worldbank.org/indicator/NY.GNP.MKTP.CD>; "Population, Total," The World Bank | Data, accessed September 25, 2020, <https://data.worldbank.org/indicator/SP.POP.TOTL>; "Budget and Economic Data," US Congressional Budget Office, January 2020, accessed September 25, 2020, www.cbo.gov/system/files/2020-01/51134-2020-01-historicalbudgetdata.xlsx; "Unraveling the Mysteries of China's Multiple Budgets," *Bloomberg*, March 13, 2020, accessed August 27, 2020, www.bloomberg.com/news/articles/2020-03-13/unraveling-the-mysteries-of-china-s-multiple-budgets-quicktake; "Budget at a Glance," Ministry of Finance, Government of India, July 5, 2019, accessed September 25, 2020, www.indiabudget.gov.in/budget2019-20/budgetglance.php; "Ethiopia's 2019–20 Budget," Cepheus Capital, July 10, 2019, accessed September 16, 2020, <https://cepheuscapital.com/wp-content/uploads/2019/01/Budget-Review-FY-2019-20.pdf>.

EXHIBIT 2A: PRICING OF SELECTED PEDIATRIC VACCINES IN THE US, 2020

Vaccine name	CDC cost/dose	Private sector cost/dose	Manufacturer
M-M-R®II: Measles, Mumps, and Rubella	\$21.71	\$78.68	Merck
TDVAX™: Tetanus and Diphtheria Toxoids	\$16.34	\$25.88	Grifols
Fluzone® Quadrivalent: Influenza	\$13.50	\$18.14	Sanofi Pasteur
Gardasil®9: Human Papillomavirus 9 Valent	\$187.01	\$227.93	Merck
Prevnar 13™: Pneumococcal 13-valent	\$143.82	\$202.00	Pfizer

Note: CDC = The Centers for Disease Control and Prevention is a national public health institute in the US.

Source: Created by the case authors based on information from “CDC Vaccine Price List,” CDC, June 25, 2020, accessed September 7, 2020, www.cdc.gov/vaccines/programs/vfc/awardees/vaccine-management/price-list/index.html.

EXHIBIT 2B: PRICING OF SELECTED VACCINES IN DEVELOPING COUNTRIES, 2018–2020

Vaccine name	Cost/dose	Manufacturer
Measles, Mumps, and Rubella	\$2.85 \$3.10	SII Sanofi Pasteur
Diphtheria, Tetanus D.T.	\$0.18 \$1.17	SII Intervax
FluQuadri®: Influenza Quadrivalent	\$2.61	Sanofi Pasteur
Human Papillomavirus Quadrivalent	\$4.50	Merck
Pneumococcal 13-Valent	\$3.30	Pfizer

Source: Created by the case authors based on information from “UNICEF Price Data Overview,” accessed July 9, 2020, https://public.tableau.com/views/UNICEFPricedataoverviewforvaccines/FullDashboard?%3Aembed=y&%3AshowVizHome=no&%3Adisplay_count=y&%3Adisplay_static_image=y&%3AbootstrapWhenNotified=true&%3Alanguage=en&:embed=y&:showVizHome=n&:apiID=host0#navType=0&navSrc=Parse; “Fluquadri Quadrivalent Influenza Vaccine,” IndiaMART, accessed September 8, 2020, www.indiamart.com/proddetail/fluquadri-quadrivalent-influenza-vaccine-20282199548.html.

EXHIBIT 3: SELECTED 2019 FINANCIAL AND OPERATIONAL DATA CONCERNING LEADING COVID-19 VACCINE MAKERS
(all figures are in USD millions unless stated otherwise)

Financial parameter ^a	SII ^b	Pfizer ^c	AstraZeneca ^d	J&J ^e	Moderna ^f
Revenues	750	39,419	24,384	42,198	60
R&D expense	-	1,047	6,059	8,834	496
Other operating expenses	392	14,579	16,942	25,572	110
Total operating expenses	392	15,626	23,001	34,406	606
EBITDA	357	23,793	2,924	7,792	-546
EBITDA/Sales	48%	60%	12%	18%	-906%
Name of COVID-19 vaccine	Covishield	BNT162b2	ChAdOx1 nCoV-19	Ad26.COV2.S	mRNA-1273
2021 estimated COVID-19 vaccine production (billions of doses)	1	1	2	1	2

Notes:

^a Business units not related to pharmaceutical or vaccine products are excluded.

^b SII fiscal year 2019, as with other Indian companies, is counted as the period from April 1, 2018 through to March 31, 2019.

^c Pfizer's Biopharmaceuticals Group, an operating segment that included the development and production of vaccines.

^d AstraZeneca's vaccine candidate was the same as Covishield.

^e J&J's Pharmaceuticals division, which excluded medical devices and consumer healthcare. Cost of Goods Sold; Selling, General & Administrative expenses were assumed at the corporate rate provided in the annual report.

^f Moderna's annual report (10K) reflected financial results from multiple products, including but not limited to vaccines.

R&D = Research and development; EBITDA = Earnings Before Interest, Taxes, Depreciation, and Amortization.

The USD/INR exchange rate is taken to be 68.915, which was the average for 2019.

The Weighted Average Cost of Capital for Indian vaccine makers can be taken to be 14%.

Source: Created by the case authors based on information from Serum Institute of India, *Standalone Financial Statements for Period 01/04/2018 to 31/03/2019*, August 26, 2019, accessed September 10, 2020; Pfizer Inc., *Form 10-K*, February 27, 2020, accessed September 10, 2020, <http://d18n0p25nwr6d.cloudfront.net/CIK-0000078003/dee171a3-b766-46e8-a807-dab4c7fb1895.pdf>; AstraZeneca, *Annual Report 2019*, February 14, 2020, accessed September 10, 2020, www.astrazeneca.com/content/dam/az/investor_relations/annual-report-2019/pdf/AstraZeneca_AR_2019.pdf; Moderna, *Annual Report Pursuant to Section 13 or 15(d) of the Securities Exchange Act of 1934 for the Fiscal Year Ended December 31, 2019*, February 27, 2020, accessed September 9, 2020, <https://investors.modernatx.com/static-files/16245ad7-c5e0-4671-82c3-a783b0c0099a>; "Technical Stock Screener, Charts, Fundamental, F&O Analysis, India," Top Stock Research, accessed September 22, 2020, www.topstockresearch.com/.

ENDNOTES

- ¹ This case has been written on the basis of published sources only. Consequently, the interpretations and perspectives presented in the case are not necessarily those of the Serum Institute of India or any of its employees.
- ² A public-private global health partnership with the goal of increasing access to immunization in poor countries. It was previously known as the Global Alliance for Vaccines and Immunization.
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¹⁰² Mayank Sen, "Serum Institute of India to Produce up to 100 Million Covid-19 Vaccine Doses for India and Low- and Middle-Income Countries as Early as 2021," Serum Institute of India, August 7, 2020, accessed September 19, 2020, www.seruminstitute.com/news_gavip_partnership_announcement.php.

¹⁰³ Kanwal, *op. cit.*

¹⁰⁴ *Ibid.*

¹⁰⁵ "SII Introduces the First-Ever Heat-Stable Rotavirus Vaccine in the World, Rotasiil," *Economic Times*, August 17, 2019, accessed September 23, 2020, <https://health.economictimes.indiatimes.com/news/pharma/sii-invents-the-first-ever-heat-stable-rotavirus-vaccine-in-the-world-rotasiil/70712802>.

¹⁰⁶ Kanwal, *op. cit.*

¹⁰⁷ Shubham Sharma, "India's Serum Institute Signs COVID-19 Vaccine Deal with America's Novavax," *Yahoo Finance*, August 6, 2020, accessed October 1, 2020, <https://in.finance.yahoo.com/news/indias-serum-institute-signs-covid-081235526.html>; "AstraZeneca Takes Next Steps Towards Broad and Equitable Access to Oxford University's Potential COVID-19 Vaccine," *op. cit.*

¹⁰⁸ Umesh Isalkar, "No Proof That MMR Vaccination Protects against Coronavirus: SII," *Times of India*, July 13, 2020, accessed October 1, 2020, <https://timesofindia.indiatimes.com/city/pune/no-proof-that-mmr-vaccination-protects-against-coronavirus-sii/articleshow/76930223.cms>.

¹⁰⁹ Himani Chandna, "How India Can Be a World Leader in Making Covid Vaccine and Keeping It Cheap Too," *The Print*, May 6, 2020, accessed September 19, 2020, <https://theprint.in/india/how-india-can-be-a-world-leader-in-making-covid-vaccine-and-keeping-it-cheap-too/414701/>.

¹¹⁰ "Covid-19 News: Oxford Trial Vaccine to be Called Covishield in India | Left, Right & Centre," *op. cit.*